
CURRICULUM CONSTRAINT TIGHTENING IN LAGRANGIAN PPO: A CONTROLLED EMPIRICAL STUDY OF λ STABILITY AND GENERALIZATION IN DISCRETE NAVIGATION

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Code: github.com/shahnawaz05/Safe_cirriculum

ABSTRACT

Lagrangian methods for constrained reinforcement learning require careful management of the dual variable λ , whose dynamics during training remain poorly understood outside continuous control benchmarks. We study whether a curriculum schedule that progressively tightens the constraint budget from 30% violation tolerance to 5% across three training stages improves λ stability and zero-shot generalization compared to standard fixed-target Lagrangian PPO. Experiments run across three random seeds on a 10×10 discrete navigation environment with four map configurations reveal two findings. The curriculum schedule produces a more stable λ trajectory, with the multiplier remaining in a narrow band (0.38–0.42) throughout training rather than climbing monotonically (0.50–0.67). Standard fixed-target Lagrangian PPO generalizes substantially better to unseen map layouts (75.3% mean success) than the curriculum variant (56.3%), despite less stable λ dynamics. The mechanism is identified: permissive early targets cause λ to decrease from its initial value, producing a persistently weaker safety signal that does not transfer as robustly. This reveals that λ stability and generalization quality are not aligned when curriculum schedules cause the multiplier to settle at a lower equilibrium. Code and trained checkpoints are available at github.com/shahnawaz05/Safe_cirriculum.

Keywords: Safe Reinforcement Learning, Constrained MDPs, Lagrangian Methods, Curriculum Learning, Proximal Policy Optimization

1 Introduction

Safe reinforcement learning addresses sequential decision-making under safety constraints, formalized as Constrained Markov Decision Processes [Altman, 1999]. Lagrangian relaxation combined with Proximal Policy Optimization [Schulman et al., 2017] has become a practical standard for this class of problems [Ray et al., 2019] due to its compatibility with existing policy gradient infrastructure. The core mechanism converts the constrained optimization into a minimax problem: a dual variable λ is updated via gradient ascent on the constraint violation, penalizing the policy whenever it exceeds the target budget. Despite its widespread adoption, the Lagrangian approach is documented to suffer from oscillatory λ dynamics [Stooke et al., 2020] that destabilize training and produce policies with unpredictable safety behavior.

Curriculum learning has been proposed as a strategy for addressing training difficulties in constrained RL. Safety-Prioritizing Curriculum Generation [Köprülü et al., 2025] generates sequences of tasks ordered by their safety demands, easing the agent into constraint satisfaction progressively. Turchetta et al. [2020] apply curriculum induction to guide safe exploration. Both lines of work operate on the task dimension: they change what the agent trains on. A distinct question has received less attention – what happens when the constraint budget itself is tightened progressively, starting with a permissive violation tolerance and reducing it toward a strict target over a single fixed task? This manipulation operates on the dual update directly rather than on task selection.

This paper presents a controlled empirical study of constraint-budget curriculum tightening applied to Lagrangian PPO in a discrete 10×10 navigation environment. Four training conditions are compared: unconstrained PPO as a performance ceiling, standard Lagrangian PPO with a fixed 5% violation target, a curriculum variant that tightens the target from 30% to 15% to 5% across three stages of 150 episodes each, and a fixed- λ ablation that removes the dual ascent update entirely. All conditions are trained on a single map configuration and evaluated zero-shot on three unseen map configurations. Experiments are conducted across three random seeds to assess consistency.

Three findings emerge from this study. **(1)** Curriculum tightening stabilizes the λ trajectory: the multiplier remains between 0.38 and 0.42 during training, compared to a monotonic rise from 0.50 to 0.67 for standard Lagrangian. **(2)** Standard Lagrangian PPO generalizes better to unseen maps by an average of 19 percentage points, despite less stable λ dynamics. **(3)** The mechanism underlying finding (2) is identified: permissive early targets cause λ to decrease during the curriculum’s first stage, resulting in a weaker final safety signal that transfers less robustly across map configurations.

2 Background

2.1 Constrained Markov Decision Processes

A Constrained Markov Decision Process (CMDP) extends the standard MDP tuple with a cost function $c : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and a scalar budget $d \geq 0$. The learning objective is

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad \text{subject to} \quad \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t c(s_t, a_t) \right] \leq d. \quad (1)$$

Lagrangian relaxation converts Equation 1 into an unconstrained minimax problem by introducing a non-negative dual variable λ :

$$\min_{\lambda \geq 0} \max_{\pi} \mathbb{E}_{\pi} \left[\sum_t \gamma^t r(s_t, a_t) \right] - \lambda \left(\mathbb{E}_{\pi} \left[\sum_t \gamma^t c(s_t, a_t) \right] - d \right). \quad (2)$$

The minimax problem in Equation 2 is solved by alternating between policy improvement steps and dual ascent on λ [Altman, 1999].

2.2 PPO-Lagrangian

PPO-Lagrangian [Ray et al., 2019] combines the clipped PPO objective with an additive constraint penalty. At each update step the policy parameters θ are updated by minimizing

$$L(\theta) = -L_{\text{CLIP}}(\theta) + \lambda \bar{C} + \alpha_v L_{\text{value}}(\theta) - \alpha_e H(\pi_{\theta}), \quad (3)$$

where L_{CLIP} is the standard PPO clipped surrogate using per-step advantages A_t , $\bar{C}$ is the mean cost over the collected episode, L_{value} is the mean-squared value error and $H(\pi_{\theta})$ is the policy entropy. The dual variable is updated after each policy update via

$$\lambda_{t+1} = \max(0, \lambda_t + \beta \cdot (v_t - d_t)), \quad (4)$$

where v_t is the smoothed violation rate over the last 20 episodes, d_t is the current constraint budget target and $\beta > 0$ is the dual learning rate. The projection onto non-negative reals ensures λ remains non-negative, since a negative multiplier would reward rather than penalize constraint violations.

Stooke et al. [2020] document that gradient ascent updates on λ tend to overshoot and oscillate, motivating PID-based extensions. An empirical study of Lagrangian methods across Safety Gymnasium tasks [Huang et al., 2025] confirms that performance is highly sensitive to the value of λ and that the optimal multiplier is task-dependent. Both observations motivate the present investigation of whether a curriculum on the budget target d_t can improve multiplier stability.

2.3 Curriculum Learning for Safe RL

Curriculum learning in safe RL has focused primarily on ordering training tasks. Turchetta et al. [2020] propose generating curricula that guide agents through progressively more constrained environments, starting from configurations where safe exploration is feasible. Köprülü et al. [2025] generate task sequences that prioritize safety early in training, showing improved sample efficiency in constrained PPO. Both methods treat the curriculum as a selector over tasks or environments, while the constraint budget remains fixed throughout training.

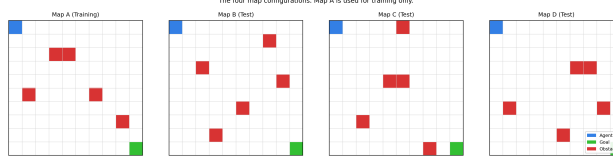


Figure 1: Map A, used for all training. Obstacles appear in red, the agent in blue and the goal in green. Maps B, C and D share the same grid size with different obstacle positions and are never seen during training.

The approach studied here is mechanistically distinct. Rather than changing the task, the budget target d_t in the dual update (Equation 4) is varied across training. A related idea appears in relaxed-exploration approaches (AAMAS 2024) that apply cost-limit schedules to CPO in continuous Safety Gymnasium environments. The present work examines the same mechanism in a discrete setting chosen for its simplicity, which allows the effect on λ dynamics to be measured in isolation from locomotion complexity.

3 Experimental Setup

3.1 Environment

Experiments use a custom discrete 10×10 grid navigation environment. At each episode the agent is placed at a randomly selected empty cell and assigned a goal at a cell at least 5 Manhattan distance units away. Five obstacle cells are fixed per map configuration. The observation is an 8-dimensional vector

$$s = \left[\frac{a_r}{10}, \frac{a_c}{10}, \frac{g_r}{10}, \frac{g_c}{10}, d_{\uparrow}, d_{\downarrow}, d_{\leftarrow}, d_{\rightarrow} \right], \quad (5)$$

where (a_r, a_c) is the agent position, (g_r, g_c) is the goal position, and each d is the number of free cells in that cardinal direction before the first obstacle or wall, normalized by the grid size. The action space is $\{0, 1, 2, 3\}$ corresponding to up, down, left and right. Moving into an obstacle or wall leaves the agent stationary. The reward is $+10$ for reaching the goal and -0.1 for every step. The cost $c_t \in \{0, 1\}$ equals 1 exactly when the attempted move would collide with an obstacle or wall. Episodes terminate at goal arrival or after 200 steps.

Four fixed map configurations are used (Figure 1). Map A serves as the sole training map. Maps B, C and D are held out and used only during zero-shot evaluation. Obstacle positions are as follows. Map A: (2,3), (2,4), (5,1), (5,6), (7,8). Map B: (1,7), (3,2), (6,5), (8,3), (4,8). Map C: (0,5), (4,4), (4,5), (7,2), (9,7). Map D: (3,6), (3,7), (6,1), (6,8), (8,5).

3.2 Network Architecture

A shared-backbone actor-critic network is used for all conditions (Figure 2). The backbone consists of two fully connected layers: Linear($8 \rightarrow 128$) followed by ReLU activation, then Linear($128 \rightarrow 64$) followed by ReLU activation. Two heads branch from the shared representation: an actor head Linear($64 \rightarrow 4$) producing raw logits from which a Categorical distribution is constructed, and a critic head Linear($64 \rightarrow 1$) producing a scalar state-value estimate $V(s)$. Advantages are computed via Generalized Advantage Estimation [Schulman et al., 2015] with $\lambda_{\text{GAE}} = 0.95$ applied to the reward signal only. Cost advantages use a zero baseline, which is standard in PPO-Lagrangian implementations [Ray et al., 2019].

3.3 Experimental Conditions

Four conditions are compared. All share identical network architecture, optimizer (Adam, learning rate 3×10^{-4}), discount $\gamma = 0.99$, GAE parameter $\lambda_{\text{GAE}} = 0.95$, PPO clip ratio $\epsilon = 0.2$, entropy coefficient $\alpha_e = 0.01$, value coefficient $\alpha_v = 0.5$, 4 gradient update epochs per episode, gradient clip norm 0.5 and a violation smoothing window of 20 episodes. Initial λ is 0.5 for all constrained conditions with $\lambda_{\text{max}} = 5.0$. The dual step size is $\beta = 0.005$ symmetrically for upward and downward updates.

Unconstrained PPO sets $\lambda = 0$ throughout training. No cost signal enters the policy gradient. This establishes a task-performance upper bound with no safety consideration.

Standard Lagrangian PPO fixes the target $d_t = 0.05$ for all 500 training episodes. The dual variable ascends whenever the smoothed violation rate exceeds 5% and descends otherwise. Starting from $\lambda_0 = 0.5$, the multiplier is free to move in either direction subject to the non-negativity constraint.

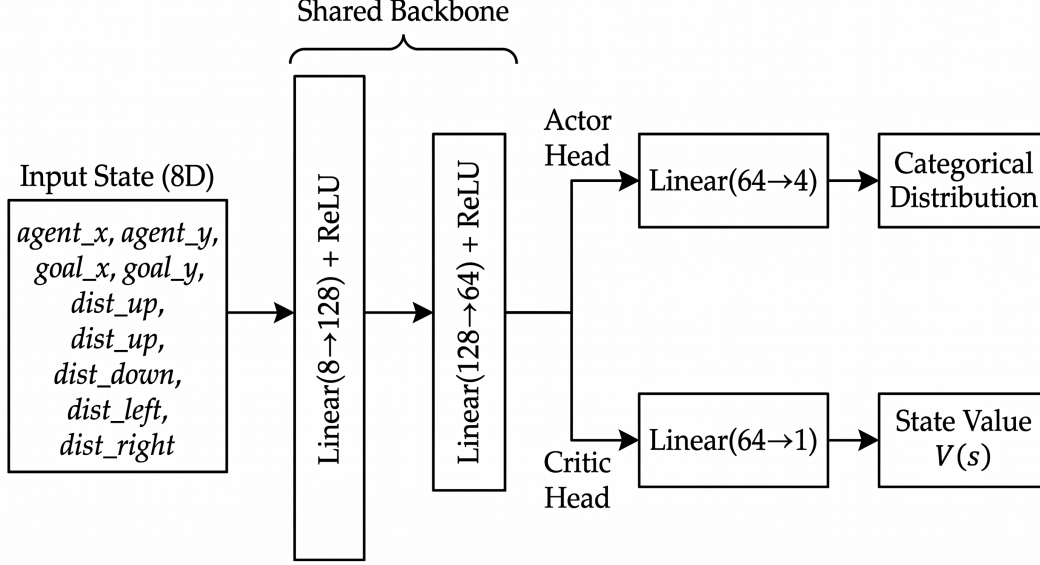


Figure 2: Actor-critic network architecture. A shared backbone processes the 8-dimensional state observation. The actor head produces action logits for the 4-action discrete space. The critic head produces a scalar value estimate used for advantage computation.

Algorithm 1 Curriculum Lagrangian PPO

- 1: Initialize policy π_θ , critic V_ϕ , $\lambda_0 = 0.5$
 - 2: Define curriculum schedule d_t per Table 1
 - 3: **for** episode $e = 1, \dots, 500$ **do**
 - 4: Collect trajectory $\tau = \{(s_t, a_t, r_t, c_t)\}$ using π_θ
 - 5: Compute reward advantages A_t via GAE on $\{r_t\}$
 - 6: Compute cost advantages G_t^c via GAE (zero baseline) on $\{c_t\}$
 - 7: Set penalized advantage $\hat{A}_t = A_t - \lambda_e G_t^c$
 - 8: **for** each gradient epoch (4 total) **do**
 - 9: Compute $L(\theta)$ via Equation 3 with $\hat{A}_t$
 - 10: Update θ via Adam; clip gradient norm at 0.5
 - 11: **end for**
 - 12: Compute smoothed violation rate v_e over last 20 episodes
 - 13: $\lambda_{e+1} = \max(0, \lambda_e + \beta(v_e - d_e))$
 - 14: **end for**
-

Curriculum Lagrangian PPO uses a three-stage tightening schedule as shown in Table 1. The dual update rule is identical to standard Lagrangian; only the target d_t changes across stages. No other hyperparameters differ between the two constrained conditions.

Fixed- λ Ablation sets $\lambda = 0.5$ throughout and disables the dual ascent update entirely. The constraint loss $\lambda \bar{C}$ still enters the total loss with a constant weight, but the weight is never adjusted. This condition isolates the effect of adaptive dual ascent by removing it.

The complete algorithm for curriculum Lagrangian PPO is given in Algorithm 1.

Table 1: Three-stage curriculum constraint tightening schedule.

Episodes	Target d_t	Rationale
1–150	0.30	Learn the task; tolerate exploration collisions
151–300	0.15	Begin tightening; policy avoids obvious obstacles
301–500	0.05	Full safety enforcement; matches standard target

Table 2: Training results on Map A. Means are computed over the final 50 episodes and averaged across 3 seeds. Standard deviations across seeds appear in parentheses.

Method	Reward	Violation	Success	Final λ
Unconstrained	-15.74 (± 0.44)	56.0% ($\pm 17.5\%$)	17.3% ($\pm 3.4\%$)	0.00
Standard Lag.	-6.56 (± 1.99)	9.0% ($\pm 1.0\%$)	62.0% ($\pm 7.5\%$)	0.67
Curriculum	-8.92 (± 3.23)	8.3% ($\pm 0.4\%$)	54.0% ($\pm 14.9\%$)	0.42
Fixed Ablation	-13.68 (± 2.09)	7.3% ($\pm 1.5\%$)	29.3% ($\pm 10.5\%$)	0.50

3.4 Evaluation Protocol

All conditions are trained on Map A for 500 episodes. Three random seeds (0, 1, 2) are used per condition, yielding 12 total training runs. Training metrics – episode reward, violation rate, success rate and λ – are recorded at every episode and aggregated across seeds for Table 2.

Zero-shot generalization is assessed using the seed 2 checkpoint for each condition, evaluated over 100 stochastic episodes on each of Maps B, C and D. Stochastic evaluation is used because it reflects the behavior of the training-time policy and avoids degenerate argmax trajectories that arise when the policy entropy is nonzero.

4 Results

4.1 Training Performance on Map A

Table 2 summarizes training outcomes averaged across 3 seeds over the final 50 episodes of each run. Values are taken directly from the seed aggregate summary.

Both Lagrangian conditions substantially reduce the violation rate relative to unconstrained PPO (9.0% and 8.3% versus 56.0%). Standard Lagrangian achieves a higher success rate (62.0%) than curriculum (54.0%) despite a comparable violation rate. The fixed ablation achieves the lowest violation rate (7.3%) but also the lowest success rate (29.3%), confirming that a static penalty without adaptive dual ascent causes excessive conservatism that prevents goal-reaching.

4.2 Lagrange Multiplier Dynamics

Figure 3 shows λ trajectories over 500 training episodes for the three non-trivial conditions.

In the standard Lagrangian condition, λ rises monotonically from 0.50 to approximately 0.67 across all three seeds, with no sign of convergence by episode 500. The consistent upward trend reflects persistent constraint violation above the 5% target throughout training.

In the curriculum condition, λ behaves in three distinct phases corresponding to the budget schedule. During Stage 1 (episodes 1–150, target $d_t = 0.30$), violation rates average approximately 15%, well below the 30% target. The dual update therefore decreases λ , bringing it from 0.50 to approximately 0.38 by episode 150. Stage 2 (target $d_t = 0.15$) continues this pattern: violations near 13% remain below the 15% target, so λ continues to decrease or holds near 0.37. At episode 301, when the target tightens to 0.05, violations (near 10%) exceed the target and λ begins rising, reaching approximately 0.42 by episode 500.

The final curriculum λ of 0.42 is 37% lower than the standard final value of 0.67. This difference in equilibrium level, not the trajectory shape, is the primary driver of the generalization gap in Section 4.4.

4.3 Training Curves

Figure 4 shows success rate and violation rate over the course of training for all four conditions, smoothed with a 25-episode moving average.

4.4 Zero-Shot Generalization to Unseen Maps

Standard Lagrangian PPO outperforms curriculum Lagrangian PPO on all three unseen maps. Table 3 reports stochastic evaluation results using seed 2 checkpoints.

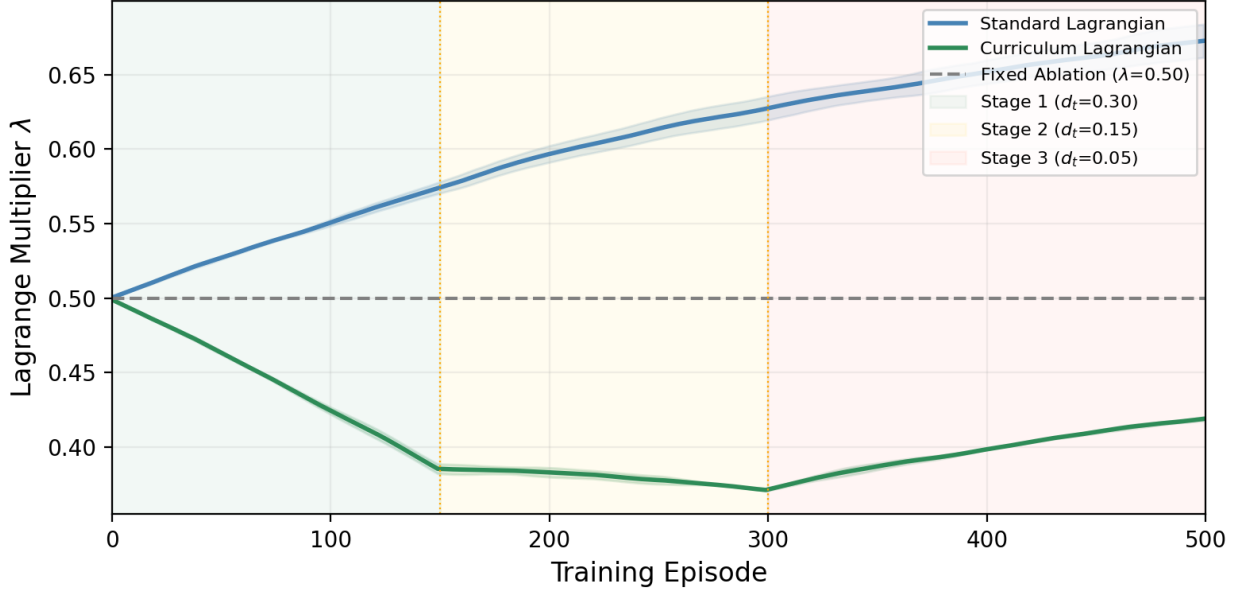
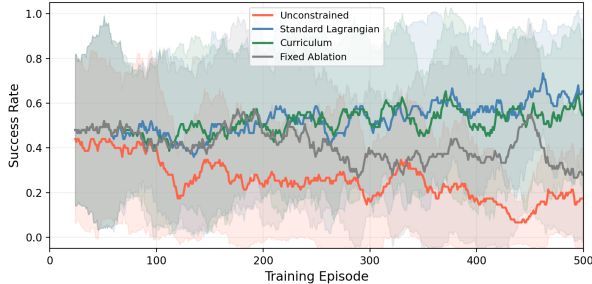
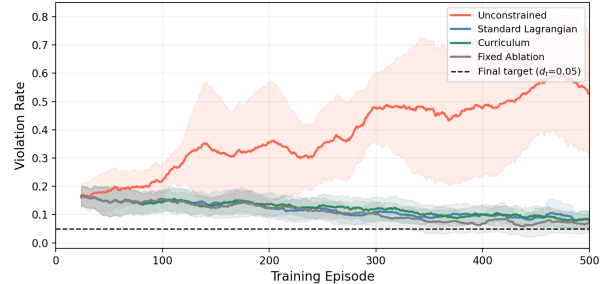


Figure 3: λ trajectories over 500 training episodes. Standard Lagrangian climbs monotonically (0.50 to 0.67). The curriculum variant decreases during permissive Stages 1 and 2, then rises in Stage 3, settling at 0.42. Vertical dashed lines mark stage transitions at episodes 150 and 300. Shaded regions correspond to the three curriculum stages.



(a) Success rate over training.



(b) Violation rate over training.

Figure 4: Training dynamics across 500 episodes for all four conditions (25-episode moving average). The dashed horizontal line in the right panel marks the final constraint target of 5%.

Standard Lagrangian achieves 75.3% mean success across the three unseen maps, compared to 56.3% for curriculum – a gap of 19 percentage points. Curriculum achieves a marginally lower collision rate (7.5% versus 8.1%), but the success rate gap is consistent across all three maps. Both constrained methods outperform unconstrained by a wide margin (19.0% mean success, 36.1% collision rate), confirming that the safety constraint mechanism transfers to unseen environments regardless of which constrained variant is used.

4.5 Ablation: Fixed λ Without Dual Ascent

The fixed- λ ablation ($\lambda = 0.5$, no update) achieves a violation rate of 7.3% on Map A – lower than both adaptive methods – but a success rate of only 29.3%. This pattern confirms that the dual ascent update is necessary for balancing safety and task performance. A fixed penalty of $0.5 \times \bar{C}$ throughout training is high enough to suppress risky behavior but too inflexible to allow the policy to take the occasional necessary risk on a path toward the goal. The adaptive update in both Lagrangian conditions provides this flexibility: the penalty adjusts as the policy improves, eventually settling at a level that enforces safety without paralysis.

Table 3: Zero-shot generalization to unseen maps. Success and collision rates from 100 stochastic evaluation episodes per map, using seed 2 checkpoints.

Method	Map B	Map C	Map D	Mean Success	Mean Collision
Unconstrained	23.0%	11.0%	23.0%	19.0%	36.1%
Standard	72.0%	74.0%	80.0%	75.3%	8.1%
Curriculum	48.0%	61.0%	60.0%	56.3%	7.5%

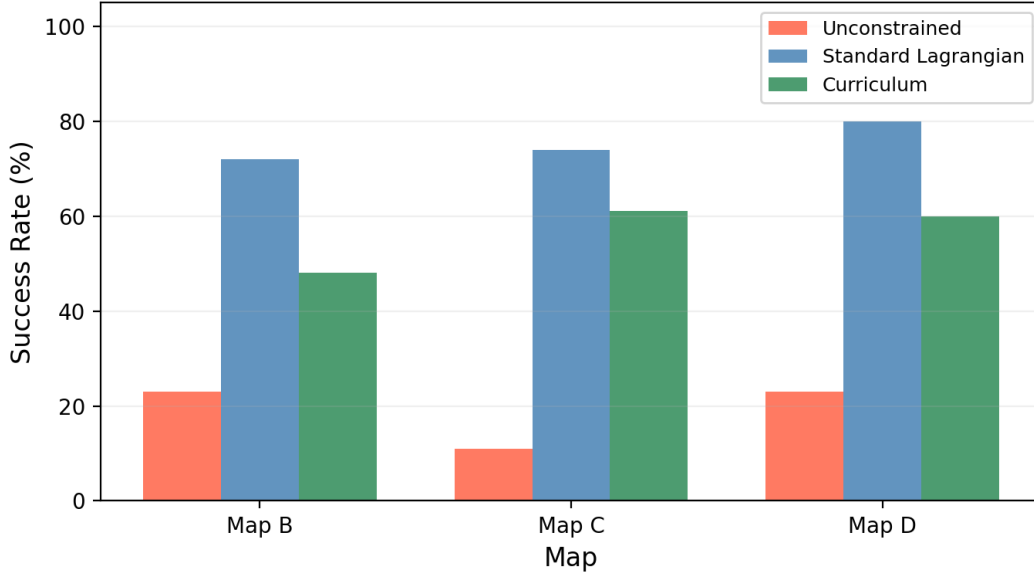


Figure 5: Zero-shot success rate on three unseen maps for all evaluated conditions. Standard Lagrangian consistently outperforms curriculum by 14–24 percentage points across all map configurations.

5 Analysis

5.1 Why the Curriculum Produces a Lower Final λ

The λ decrease during Stage 1 of curriculum training is a direct consequence of the dual update rule. When $v_t < d_t$, the update $\lambda \leftarrow \max(0, \lambda + \beta(v_t - d_t))$ reduces λ . In Stage 1 the target is $d_t = 0.30$, but the actual violation rate is approximately 0.15 across all seeds. The gap is $v_t - d_t \approx -0.15$, and with $\beta = 0.005$ the update reduces λ by roughly 0.00075 per episode, or about 0.11 over 150 episodes. Starting from $\lambda_0 = 0.5$, this brings λ to approximately 0.39 by episode 150, consistent with the observed trajectories.

Stage 2 continues this pattern (violations near 0.13 against target 0.15), preventing recovery. Only in Stage 3, when the target tightens to 0.05 and violations are near 0.10, does λ begin climbing. However, 200 episodes at a rate of +0.00025 per episode yields a recovery of only about 0.05. The final curriculum λ of approximately 0.42 reflects this arithmetic precisely.

Standard Lagrangian, facing a tight 0.05 target from episode 1 against initial violations of 0.15, sees $\Delta = 0.10$ at every update and climbs at +0.0005 per episode. Over 500 episodes this yields an increase of approximately 0.25, bringing λ from 0.50 toward 0.75. The observed value of 0.67 is somewhat lower, consistent with downward corrections as the policy improves mid-training.

The consequence is that the curriculum agent receives a safety penalty of $0.42 \times \bar{C}$ at episode 500, while standard receives $0.67 \times \bar{C}$ – a 60% stronger signal. This stronger signal appears to drive deeper internalization of obstacle-avoidance behavior during training, which then transfers more robustly when obstacle positions change at test time.

5.2 Implications for Curriculum Design

The finding suggests that curriculum constraint tightening must account for the dual update’s behavior during permissive phases. When the initial target is set substantially above the agent’s natural violation rate, the dual variable decreases – undermining the curriculum’s intent. Two design choices could prevent this. First, initializing λ at a lower value (e.g., 0.1 instead of 0.5) for the curriculum condition would reduce the magnitude of the decrease during Stage 1. Second, using an asymmetric dual update rate – slower decrements during permissive stages and faster increments during tight stages – would preserve more of the initial safety signal.

A more fundamental alternative is to decouple the curriculum schedule from the dual update during permissive stages: hold λ fixed during Stages 1 and 2, and activate the dual update only when the tight target is enforced in Stage 3. This would preserve the exploration benefit of the loose early constraint while ensuring the final safety signal matches or exceeds that of standard Lagrangian.

Empirical studies in safe RL that identify failure modes of intuitive approaches are equally valuable as those confirming benefits. The present finding – that λ stability and generalization quality are decoupled when curriculum schedules cause multiplier relaxation – is a specific, reproducible and mechanistically explained failure mode. This characterization can inform the design of adaptive curriculum schedules, the choice of λ initialization and the interpretation of λ stability metrics in future work.

6 Limitations

This study has five limitations that bound the scope of its conclusions.

Single environment. All experiments use a 10×10 discrete gridworld. Whether the identified mechanism operates the same way in continuous control environments such as Safety Gymnasium remains to be established. The discrete setting was chosen for analytical clarity, not as a claim about generality.

Hand-designed schedule. The curriculum schedule has stage boundaries fixed at episodes 150 and 300. An adaptive schedule that advances stages based on measured violation rates rather than episode counts may produce different results. The comparison reported here is between two specific fixed schedules.

Limited seeds. Results are based on three random seeds. While observed patterns are consistent across seeds, the standard deviation of the curriculum success rate (14.9%) indicates meaningful variability that a larger seed count would resolve.

Fixed dual step size. The step size $\beta = 0.005$ was selected based on preliminary experimentation and held constant across all conditions. The sensitivity of the findings to this parameter was not systematically studied.

Single-seed generalization. Zero-shot generalization is evaluated using the seed 2 checkpoint rather than averaging over checkpoints from all seeds. This was a practical decision given evaluation time, but it introduces additional variance in the reported generalization numbers.

7 Conclusion

A controlled empirical study compared four Lagrangian PPO variants in a discrete 10×10 navigation environment, focusing on the effect of a three-stage constraint-budget curriculum on multiplier stability and zero-shot generalization across unseen map configurations.

Two findings emerge. The curriculum schedule stabilizes the λ trajectory, keeping the multiplier in a narrow band (0.38–0.42) relative to the monotonic climb observed in standard Lagrangian (0.50–0.67). At the same time, standard Lagrangian generalizes better to unseen maps by an average of 19 percentage points. The identified mechanism – permissive early targets causing λ to decrease, producing a persistently weaker safety signal – explains why the two desirable outcomes are not aligned in this setting.

Future work should investigate curriculum schedules that control for the final λ level, for instance by using asymmetric dual update rates or by activating the dual ascent only during the strict final stage. Replicating the study in continuous control environments would establish whether the mechanism operates similarly under locomotion complexity. The discrete gridworld studied here provides a clean reference point against which such extensions can be compared.

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